

C++ Course

Materials for Students

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Chapter 1

Introduction

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This ebook is the first English draft based on the Polish C++ learning materials. The Polish version remains the source of truth for now. The English version starts with the course structure, learning path, and translation plan.

The full course is organized from environment setup and C++ basics to functions, arrays, strings, files, exceptions, object-oriented programming, STL, build systems, testing, and final projects.

Chapter 2

English Version Plan

2.1 Goal

The English edition should become a downloadable version of the C++ course for students who want to learn from the same material as the Polish course.

The first release should focus on readability and navigation. The code examples can stay close to the Polish repository at the beginning, while explanations, chapter introductions, exercises, and project descriptions are translated in batches.

2.2 Learning Path

1. `00-start` - development environment, compiler, editor, Git, and CMake.
2. `01-basics` - first program, input/output, types, variables, and operators.
3. `02-control-flow-loops` - conditions, `switch`, loops, `break`, `continue`.
4. `03-functions-arrays-strings` - functions, arrays, matrices, and strings.
5. `04-pointers-references-memory` - addresses, pointers, references, and safe memory habits.
6. `05-files-exceptions` - text files, validation, and exceptions.
7. `06-oop` - classes, objects, constructors, encapsulation, inheritance, and operators.
8. `07-stl-data-structures` - STL containers, algorithms, maps, stacks, queues, and lists.
9. `08-project-build-tests` - multi-file projects, build scripts, tests, and CI.
10. `09-modern-cpp` - selected modern C++ features, lambdas, move semantics, namespaces, and threads.
11. `10-projects` - final project topics, grading checklists, and project planning.

2.3 Translation Batches

2.3.1 Batch 1

- Main repository introduction.
- Ebook front matter.
- `00-start`.
- `01-basics`.

2.3.2 Batch 2

- `02-control-flow-loops`.
- Bridge exercise: minimum, maximum, and average.

2.3.3 Batch 3

- `03-functions-arrays-strings`.
- Bridge exercise: GADERYPOLUKI cipher.

2.3.4 Batch 4

- 04-pointers-references-memory.
- 05-files-exceptions.
- 06-oop.
- Bridge exercise: figures and polymorphism.

2.3.5 Batch 5

- 07-stl-data-structures.
- 08-project-build-tests.
- Bridge exercise: RPN calculator.

2.3.6 Batch 6

- 09-modern-cpp.
- 10-projects.
- Project checklist and grading card.

2.4 Editorial Rules

- Keep the English text simple and practical.
- Do not translate code identifiers unless the example itself is rewritten.
- Prefer short paragraphs and clear section titles.
- Keep exercises close to the Polish version, but adjust wording for an English-speaking student.
- Do not include archive materials in the ebook.
- Keep the Polish and English ebook versions buildable by the same script.

2.5 YouTube Support

The first bilingual video format should be short quizzes:

- Polish question first.
- English question second.
- One C++ concept per video.
- Link to the repository and both ebook downloads in the description.

Example:

C++ Quiz #1

What will this program print?

```
int x = 5;
std::cout << x++ << " " << x;
```

- A) 5 5
- B) 5 6
- C) 6 6

Answer: B) 5 6, because postfix `x++` returns the current value first and increments the variable afterwards.

2.6 First Downloadable English Draft

The first downloadable English file should be marked clearly as a draft:

- title: C++ Course
- version: 0.1.0-en
- scope: plan, learning path, and translation roadmap
- output files:

- `ebook/download/cpp-course-en.pdf`
- `ebook/download/cpp-course-en.epub`